

Managing common neurological complications following cardiac surgery.

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Despite recent technological advances in the field of surgery, neurological complications remain a significant post-operative issue. Two of the most common post-operative complications are delirium and post-operative cognitive decline, which occur frequently after cardiac surgery. This article provides an overview of delirium and post-operative cognitive decline, including risk factors, signs and symptoms, and diagnosis. The author also considers the nurse's role in managing patients who are at risk of, or have experienced, delirium or post-operative cognitive decline after cardiac surgery.

Risk factors for postoperative delirium in patients after coronary artery bypass grafting: A prospective cohort study.

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This study was designed to identify the incidence and independent perioperative risk factors associated with postoperative delirium of patients who underwent coronary artery bypass grafting (CABG) in a large intensive care unit setting in China. Delirium was diagnosed by the confusion assessment method for the intensive care unit (CAM-ICU). Baseline demographics, perioperative data, and postoperative outcomes of 249 consecutive patients who underwent CABG were recorded prospectively and analyzed via univariate analysis and multivariate logistic regression to determine the independent risk factors of postoperative delirium. Postoperative delirium was detected in 76 patients according to CAM-ICU criteria. The incidence was 30.52%. Patients with and without delirium differed significantly on 34 variables ($P < .05$). Multivariate logistic regression analysis revealed that preoperative atrial fibrillation (odds ratio [OR], 3.957; 95% confidence interval [CI], 1.727-9.066), elevated European system for cardiac operative risk evaluation (OR, 1.178; 95% CI, 1.018-1.364), cognitive impairment (OR, 3.231; 95% CI, 1.008-10.356), prolonged surgery duration (OR, 1.008; 95% CI, 1.003-1.014), postoperative poor quality of sleep (OR, 5.001; 95% CI, 2.476-10.101), and electrolyte disturbance (OR, 2.095; 95% CI, 1.041-4.216) were independently associated with postoperative delirium after CABG. Delirium is a frequent complication. Factors independently associated with delirium are preoperative atrial fibrillation, elevated European system for cardiac operative risk evaluation and cognitive impairment, longer surgery duration, postoperative poor quality of sleep, and electrolyte disturbance. The study may be helpful in decreasing the incidence of postoperative delirium after CABG by treating these predictors properly.

Delirium is a Common and Early Finding in Patients in the Pediatric Cardiac Intensive Care Unit.

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To determine incidence, associated risk factors, and characteristics of delirium in a pediatric cardiac intensive care unit (CICU). Delirium is a frequent and serious complication in adults after cardiac surgery, but there is limited understanding of its impact in children with critical cardiac disease. Single-center prospective observational study of CICU patients ≤ 21 years old. All were screened for delirium using the Cornell Assessment for Pediatric Delirium each 12-hour shift. Ninety-nine patients were included. Incidence of delirium was 57%. Median time to development of delirium was 1 day (95% CI 0, 1 days).

Children with delirium were younger (geometric mean age 4 vs 46 months; $P < .001$), had longer periods of mechanical ventilation (mean 35.9 vs 8.8 hours; $P = .002$) and had longer cardiopulmonary bypass times (geometric mean 126 vs 81 minutes; $P = .001$). Delirious patients had longer length of CICU stay than those without delirium (median 3 (IQR 2, 12.5) vs 1 (IQR 1, 2) days; $P < .0001$). A multivariable generalized linear mixed model showed a significant association between delirium and younger age (OR 0.35 for each additional month, 95% CI 0.19, 0.64), need for mechanical ventilation (OR 4.1, 95% CI 1.7, 9.89), and receipt of benzodiazepines (OR 3.78, 95% CI 1.46, 9.79). Delirium is common in patients in the pediatric CICU and is associated with longer length of stay. There may be opportunities for prevention of delirium by targeting modifiable risk factors, such as use of benzodiazepines.

Impact of dexmedetomidine on the incidence of delirium in elderly patients after cardiac surgery: A randomized controlled trial.

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Delirium is a frequent complication after cardiac surgery and its occurrence is associated with poor outcomes. The purpose of this study was to investigate the impact of perioperative dexmedetomidine administration on the incidence of delirium in elderly patients after cardiac surgery. This randomized, double-blinded, and placebo-controlled trial was conducted in two tertiary hospitals in Beijing between December 1, 2014 and July 19, 2015. Eligible patients were randomized into two groups. Dexmedetomidine (DEX) was administered during anesthesia and early postoperative period for patients in the DEX group, whereas normal saline was administered in the same rate for the same duration for patients in the control (CTRL) group. The primary endpoint was the incidence of delirium during the first five days after surgery. Secondary endpoints included the cognitive function assessed on postoperative days 6 and 30, the overall incidence of non-delirium complications within 30 days after surgery, and the all-cause 30-day mortality. Two hundred eighty-five patients were enrolled and randomized. Dexmedetomidine did not decrease the incidence of delirium (4.9% [7/142] in the DEX group vs 7.7% [11/143] in the CTRL group; OR 0.62, 95% CI 0.23 to 1.65, $p = 0.341$). Secondary endpoints were similar between the two groups; however, the incidence of pulmonary complications was slightly decreased (OR 0.51, 95% CI 0.26 to 1.00, $p = 0.050$) and the percentage of early extubation was significantly increased (OR 3.32, 95% CI 1.36 to 8.08, $p = 0.008$) in the DEX group. Dexmedetomidine decreased the required treatment for intraoperative tachycardia (21.1% [30/142] in the DEX group vs 33.6% [48/143] in the CTRL group, $p = 0.019$), but increased the required treatment for postoperative hypotension (84.5% [120/142] in the DEX group vs 69.9% [100/143] in the CTRL group, $p = 0.003$). Dexmedetomidine administered during anesthesia and early postoperative period did not decrease the incidence of postoperative delirium in elderly patients undergoing elective cardiac surgery. However, considering the low delirium incidence, the trial might have been underpowered. ClinicalTrials.gov NCT02267538.

Stroke, Seizures, Hallucinations and Postoperative Delirium as Neurological Complications after Cardiac Surgery and Percutaneous Valve Replacement.

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(1) Background: Neurological complications such as acute ischemic stroke or postoperative delirium are frequent after cardiac surgery or percutaneous valve replacement. This study aimed to identify

corresponding risk factors. (2) Methods: 297 patients with percutaneous valve replacement or cardiac surgery were postoperatively screened for neurological complications such as delirium, stroke, seizures and hallucinations twice daily for three days. Pre- and perioperative risk factors were evaluated in a multivariate model. (3) Results: Neurological complications occurred in 43.8% (n = 130) as composed of delirium (43.43%, n = 129), stroke (2.7%, n = 8), seizures (1.35%, n = 4) and real hallucinations (3.36%, n = 10). Multiple logistic regression revealed an association of neurological complications with lower Montreal Cognitive Assessment scores (Exp(B) 2.042; 95% CI, 1.183–3.525, p = 0.010), older age (Exp(B) 1.071; 95% CI, 1.036–1.107, p < 0.001), red blood cell transfusions until postoperative day 3 (Exp(B) 1.157; 95% CI, 1.030–1.300, p = 0.014), history of heart failure (Exp(B) 1.985; 95% CI, 1.130–3.487, p = 0.017) and increased CRP levels (Exp(B) 1.004; 95% CI, 1.000–1.008, p = 0.037). (4) Conclusions: Postoperative delirium remains a frequent complication after cardiac surgery, while stroke and seizures occur rarely. A preoperative risk profile including older age, history of heart failure and cognitive impairment was identified for a complicated postoperative course. However, the impact of an intense inflammatory response must not be neglected.

Perioperative dexmedetomidine reduces delirium after coronary artery bypass graft surgery: A prospective, single-blind, observational study.

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Delirium is a commonly seen complication of cardiac surgery. Dexmedetomidine, by its anti-inflammatory properties and other effects, can attenuate postoperative delirium. The aim of this work was to study the incidence of delirium after coronary artery bypass graft surgery, and to compare the effects of dexmedetomidine and propofol on the incidence of postoperative delirium in coronary artery bypass graft surgery patients. A prospective, observational study was conducted on 180 consecutive patients undergoing off-pump or on-pump coronary artery bypass graft surgery. The patients were administered either intravenous dexmedetomidine (n = 90) or propofol (n = 90) after hemostasis was achieved, till they were ready for weaning from the ventilator. The Confusion Assessment Method was used to assess the incidence of postoperative delirium. A total of 25 (13.8%) patients developed delirium after coronary artery bypass graft surgery. Sedation with dexmedetomidine was associated with a significantly reduced incidence of postoperative delirium (8.9% v 18.9% propofol, P = 0.049). Subgroup analyses showed reduced incidence of postoperative delirium in off-pump patients compared to on-pump coronary artery bypass graft patients (3.3% vs. 20%, P = 0.009 dexmedetomidine group and 11.6% vs. 33.3%, P = 0.047 propofol group respectively). The mean age of the patients who had delirium was significantly more (64.9 ± 8.1 years vs. 52.5 ± 5.8 years, P = 0.046) compared to those who did not have delirium. Administration of dexmedetomidine-based sedation resulted in the reduced incidence of postoperative delirium compared to propofol-based sedation in patients after coronary artery bypass graft surgery.

Early post-cardiac surgery delirium risk factors.

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The purpose of this study was to identify the post-cardiac surgery delirium risk factors and to evaluate clinical outcomes. Data on 90 patients with postoperative delirium after cardiac surgery on cardiopulmonary bypass (CPB) were analyzed retrospectively. The patients were divided into two groups

by evaluating the severity of the delirium: light and moderate delirium group (n=74) and severe delirium group (n=16). We found that the rate of early post-cardiac surgery delirium was low (4.17%). We have determined that post-cardiac surgery delirium prolonged the length of stay in the Intensive Care Unit (ICU) by (8.4 (8.6)) and the hospital stay by (23.6 (13.0)) days. The patients had higher preoperative risk scores, their age was 71.5 (8.9) years, the body mass index was 28.8 (4.4) kg/m², the majority were male (72.2%), and the left ventricular ejection fraction was 46.1(11.9) %. Statistical analysis by multivariable logistic regression has indicated that increasing the dose of fentanyl administered during surgery over 1.4 mg also increased the possibility of developing a severe delirium (OR=29.4, CI 4.1-210.3) and longer aortic clamping time could be independently associated with severe postoperative delirium (OR=8.0, CI 1.7-37.2). After surgery, new atrial fibrillation (AF) episodes amounted to 53.3% and, after distinguishing the delirium severity groups, AF developed in the patients belonging to the severe delirium groups statistically significantly more frequently, 81.8 vs 47.3, where p=0.01. Our data suggest that early post-cardiac surgery delirium is not a common complication, but it prolonged the length of stay at the ICU and in the hospital. The delirium risk factors, such as longer aortic clamping time, the dose of fentanyl and new atrial fibrillation episodes occurring after cardiac surgery, are associated statistically significantly with the development of severe post-cardiac surgery delirium.

ICU Delirium in Cardiac Patients.

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Delirium is a prevalent complication in critically ill medical and surgical cardiac patients. It is associated with increased morbidity and mortality, prolonged hospitalizations, cognitive impairments, functional decline, and hospital costs. The incidence of delirium in cardiac patients varies based on the criteria used for the diagnosis, the population studied, and the type of surgery (cardiac or not cardiac). Delirium experienced when cardiac patients are in the intensive care unit (ICU) is likely preventable in most cases. While there are many protocols for recognizing and managing ICU delirium in medical and surgical cardiac patients, there is no homogeneity, nor are there established clinical guidelines. This review provides a comprehensive overview of delirium in cardiac patients and highlights its presentation, course, risk factors, pathophysiology, and management. We define cardiac ICU patients as both medical and postoperative surgical patients with cardiac disease in the ICU. We also highlight current controversies and future considerations of innovative therapies and nonpharmacological and pharmacological management interventions. Clinicians caring for critically ill patients with cardiac disease must understand the complex syndrome of ICU delirium and recognize the impact of delirium in predicting long-term outcomes for ICU patients.

Systematic literature review on early detection of postoperative delirium in adult patients after cardiac surgery.

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Postoperative delirium (POD) is a cognitive decline and attention deficit that can occur in patients after cardiac surgery. Despite extensive research identifying the risk factors, POD often remains undiagnosed and untreated in medical settings. Therefore, this systematic literature review (SLR) aimed to summarize the available studies on early POD identification in patients following cardiovascular surgery. Data were

obtained from Scopus, PubMed, and the Cochrane International database. The search strategy was designed to identify articles published between January 2000 and January 2024. The keywords and medical subject heading (MeSH) terms used in the search included "postoperative delirium," "cardiac surgery," "early detection," and "risk factors." Studies were included if they met the following criteria: (1) focused on adult patients undergoing cardiac surgery; (2) examined methods for the early identification or prediction of POD; and (3) provided statistical analyses or clinical outcomes related to POD. The exclusion criteria were: studies on non-cardiac surgeries, pediatric populations, or those without clear methodological details. Seven of the examined studies highlighted high-risk variables (individual characteristics, underlying diseases, and treatment methods) and typical symptoms as crucial components for early POD diagnosis. However, the diagnostic criteria and evaluation methodologies needed to be more consistent, and there was no consensus regarding the most efficient early detection approaches. These findings underscore the need for standardized diagnostic criteria and evaluation methods for the early detection of POD in patients undergoing cardiovascular surgery. Evidence-based recommendations are necessary to enhance early diagnosis and treatment of POD in this population.

Risk factors of delirium after cardiac surgery: a systematic review and meta-analysis.

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Post-operative delirium (POD) is a relatively common occurrence following surgical procedures, particularly cardiac surgeries. Given that the majority of pharmacologic treatments for delirium have demonstrated inadequate efficacy, it is of great importance to identify risk factors to prevent delirium or reduce its complications. Consequently, in this systematic review and meta-analysis, we identified risk factors of POD after cardiac surgery. A comprehensive search of the literature was conducted using the databases Scopus, PubMed, and Web of Science from the inception to April 22, 2024. The objective was to identify prospective cohorts that had assessed the risk factors associated with POD in patients undergoing cardiac surgery using multivariate regression. Of the 3,166 studies that were initially screened, 23 were included in the review. Nine risk factors were evaluated including age (OR 1.06, 95% CI (1.04, 1.08), $p < 0.001$), pre-operative depression (OR 3.71, 95% CI (2.45, 5.62), $p < 0.001$), post-operative atrial fibrillation (AF) (OR 2.39, 95% CI (1.79, 3.21), $p < 0.001$), hypertension (HTN) (OR 1.64, 95% CI (0.75, 3.56), $p = 0.212$), age ≥ 65 (OR 3.32, 95% CI (2.40, 4.60), $p < 0.001$), pre-operative AF (OR 4.43, 95% CI (2.56, 7.69), $p < 0.001$), diabetes mellitus (OR 2.16, 95% CI (1.39, 3.35), $p = 0.001$), combined coronary artery bypass graft (CABG) + valve surgery (OR 2.73, 95% CI (1.66, 4.49), $p < 0.001$), and cardiopulmonary bypass (CPB) time (OR 1.02, 95% CI (1.01, 1.04), $p = 0.001$). A total of nine risk factors were evaluated, from which eight were found to have a statistically significant effect on the risk of developing POD. These factors can be employed to more effectively identify at-risk patients and to prevent the occurrence of POD. Furthermore, this approach can facilitate earlier diagnosis and more effective patient care.